Dmytro Palahin

Quantitative Researcher / Quantitative Engineer

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Final-year Computer Science engineering student with **3 years of experience in Data Engineering, Machine Learning and MLOps** at **Société Générale Assurances**. Strong background in **applied mathematics, probability, statistics, optimization and algorithms**, complemented by hands-on experience in **scientific computing and large-scale data processing**. Interested in **quantitative finance, systematic trading and data-driven strategies**, with a focus on **quantitative research, statistical modeling and market data analysis**.

PROFESSIONAL EXPERIENCE

Data Engineer / MLOps (Work-study) — Société Générale Assurances **09/2023 – Present**

La Défense, France

- Analyzed **1 year of operational data** from the **Zaion Callbot** using Python to identify the main sources of errors, assess system performance, and formulate **3 improvement recommendations**
  - *Python · Pandas · NumPy · Jupyter Notebook · Matplotlib · Plotly*
- Designed an **automated SAS Guide-to-Python migration system**, integrating data processing with **Polars and DuckDB**, prompt engineering, and **automated validation through a 2-level test suite**
  - *Python · Polars · DuckDB · GitHub Copilot · AWS S3*
- Designed and automated a **data processing and migration pipeline** enabling the transfer of **500+ GB of SAS Guide datasets to AWS S3** and making them available to Data teams
  - *Python · AWS S3 · fsspec · smbfs · Linux · Bash*

TECHNICAL & SCIENTIFIC PROJECTS

- **Combinatorial Optimization of Logistics Centers** | *Python, Julia, CPLEX, MILP* **2025**
  Modeling and solving a facility location problem using **mixed-integer linear programming**, CPLEX and heuristic methods
- **Scientific Publication — Mathematical Algorithms** **2021**
  Development of a mathematical method for impulse noise filtering in video images, published in *Informatics and Mathematical Methods in Simulation*, Vol. 11, No. 4
- **Anomaly Detection** | *Python, Machine Learning* **2021**
  Study of anomaly detection methods and partial data labeling
- **Traveling Salesman Problem** | *C, Combinatorial Optimization, Metaheuristics* **2023**
  Implementation in C of an optimization algorithm based on **ant colonies**
- **Machine Learning Phishing Detection** | *Python, Machine Learning, NLP* **2026**
  Classification of **76 677 emails** from 5 real-world datasets using TF-IDF, LightGBM, Naive Bayes et LSTM

EDUCATION

Sup Galilée Engineering School — Engineering Degree in Computer Science *(in progress)* **2023 – 2026**

Courses : Probability and Statistics, Linear and Combinatorial Optimization, Numerical Methods, Data Mining, Programming for AI, Algorithms and Complexity, Graph Algorithms, C/C++, Parallel and Distributed Algorithms, Advanced Databases, Cloud Computing

Sorbonne Paris Nord University — Preparatory Class for Sup Galilée **2021 – 2023**

Courses : Linear Algebra (vector spaces, matrices, eigenvalues, diagonalization), Multivariable Analysis (partial derivatives, Jacobian, Hessian, Taylor series), Probability (random variables, conditional probability, common distributions), differential equations and numerical computing

CERTIFICATIONS & SKILLS

- **Programming & Scientific Computing :** C/C++, Python, Julia, Matlab, SQL, SAS Guide, Java, Haskell · NumPy, Pandas, Polars, DuckDB · Git, Linux, Jupyter
- **Algorithms & Computer Science :** Data structures, algorithmic complexity, graph algorithms, parallel and distributed programming
- **Applied Mathematics :** Linear algebra, probability and statistics, numerical methods, optimization, Monte Carlo simulation
- **Machine Learning :** Scikit-Learn, LightGBM, XGBoost, TensorFlow/Keras · Data Mining, classification, anomaly detection
- **Financial Markets & Market Data :** Forex, equities, futures · Market microstructure, order book (L2/L3), market data, liquidity, order types, execution, VWAP · COT reports (Commitments of Traders), risk management
- **Udemy :** *Python for Data Science and Machine Learning Bootcamp* — Jose Portilla / Pierian Training

AWARDS & PARTICIPATIONS

- **Georges Besse Foundation Laureate**
- **Eco – Techno Ukraine ISEF** — 1st place in Computer Science & **Collective Intelligence Internet Service** — 1st place in Ukraine

LANGUAGES & INTERESTS

French : Fluent **English :** Fluent **Ukrainian :** Native **Russian :** Bilingual

Financial Markets / Trading · Chess · Ski · Tennis · Swimming · Karting | Formula 1